Lecture Activity 2

IT325 Machine Learning | EDA Planning for the Performance Innovative Task

Type	Lecture Activity
Topic Focus	Introduction to exploratory data analysis for text classification
Time Allotment	45 to 60 minutes
PIT Connection	Students inspect and plan the EDA phase of their approved dataset
Mode	Individual classroom activity

Learning Outcomes

1. Identify the text column and target variable in an approved PIT dataset.
2. Describe the size, structure, and basic quality of the dataset.
3. Plan suitable EDA steps before model training begins.

Activity

In this activity, students will examine their approved text classification dataset and prepare an EDA plan. The goal is to understand the problem, the available variables, and the quality of the data before any preprocessing or model training is performed.

Materials Needed

1. Approved dataset link or CSV file
2. Notebook, worksheet, or digital document for responses
3. Laptop for quick dataset inspection if available

Classroom Procedure

1. The instructor gives a short discussion on why EDA is necessary in machine learning projects.
2. Students review the PIT requirements and confirm that their dataset matches the approved topic and format.
3. Students inspect the dataset and answer the guide questions individually.
4. Selected students share their observations and proposed EDA steps.

Expected Output

A one page EDA planning sheet or short written response that describes the dataset, the target variable, possible data quality issues, and the EDA tasks to be done in the laboratory.

Questions

1. What is the title or source of your dataset?

The title of the dataset is Stroke Prediction, it is sourced on Kaggle.
https://www.kaggle.com/datasets/teamincribo/stroke-prediction?select=stroke_prediction_dataset.csv

2. What real world problem does your dataset help solve?

It helps healthcare providers predict the risk of a stroke based on a patient's symptoms, lifestyle habits (smoking/alcohol), and clinical metrics (BMI, Glucose levels) to allow for early preventive treatment.

3. Which column contains the text data?

Symptoms

4. What is your target variable?

Diagnosis

5. How many rows and columns are in the dataset?

15,000 rows and 22 columns

6. Are there missing values, duplicates, or columns that may not be useful?

There are 2,500 missing values in Symptoms column. There are no duplicates and the columns that are not useful are the Patient ID and Patient Name.

7. What EDA steps will you perform before training models?

1. Handle the 2,500 missing values in "Symptoms" by labeling them as "No Symptoms Reported."
2. Create a heatmap to see how physical factors like Age and Average Glucose Level relate to a stroke diagnosis.
3. Analyze the frequency of specific words in the Symptoms column to see which specific symptoms have the highest correlation with a positive stroke diagnosis.
4. Verify that the number of "Stroke" vs "No Stroke" cases is balanced to ensure the model isn't

biased.

Screenshot of your code

```
import pandas as pd

file_path = 'data/stroke_prediction_dataset.csv'
df = pd.read_csv(file_path)

print("Stroke Prediction Dataset")

print(f"\nDataset Size:")
print(f"Rows: {df.shape[0]}")
print(f"Columns: {df.shape[1]}")

print(f"\nData Quality Check:")
print(f"Missing Values (Symptoms): {df['Symptoms'].isnull().sum()}")
print(f"Duplicates found: {df.duplicated().sum()}")

print(f"\nClass Distribution:")
print(df['Diagnosis'].value_counts())


print(df[['Symptoms', 'Diagnosis', 'Age', 'Average Glucose Level']].head())
```

```
Stroke Prediction Dataset

Dataset Size:
Rows: 15000
Columns: 22

Data Quality Check:
Missing Values (Symptoms): 2500
Duplicates found: 0

Class Distribution:
Diagnosis
No Stroke    7532
Stroke       7468
Name: count, dtype: int64
                                            Symptoms  Diagnosis  Age  \
0                      Difficulty Speaking, Headache     Stroke   56
1    Loss of Balance, Headache, Dizziness, Confusion     Stroke   80
2                                Seizures, Dizziness     Stroke   26
3  Seizures, Blurred Vision, Severe Fatigue, Head...  No Stroke   73
4                                Difficulty Speaking     Stroke   51

   Average Glucose Level
0                 130.91
1                 183.73
2                 189.00
3                 185.29
4                 177.34
```